Dynamic and Adaptive Self-Training for Semi-Supervised Remote Sensing Image Semantic Segmentation

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***Abstract*— Remote sensing (RS) technology has made remarkable progress, providing a wealth of data for various applications, such as ecological conservation and urban planning. However, the meticulous annotation of this data is labor-intensive, leading to a shortage of labeled data, particularly in tasks like semantic segmentation. Semi-supervised methods, combining consistency regularization (CR) with self-training, offer a solution to efficiently utilize labeled and unlabeled data. However, these methods encounter challenges due to imbalanced data ratios. To tackle these challenges, we introduce a self-training approach named dynamic and adaptive self-training (DAST), which is combined with dynamic pseudo-label sampling (DPS), distribution matching (DM), and adaptive threshold updating (ATU). DPS is tailored to address the issue of class distribution imbalance by giving priority to classes with fewer samples. Meanwhile, DM and ATU aim to reduce distribution disparities by adjusting model predictions across augmented images within the framework of CR, ensuring they align with the actual data distribution. Experimental results on the Potsdam and iSAID datasets demonstrate that DAST effectively balances class distribution, aligns model predictions with data distribution, and stabilizes pseudo-labels, leading to state-of-the-art performance on both datasets. These findings highlight the potential of DAST in overcoming the challenges associated with significant disparities in labeled-to-unlabeled data ratios.**

***Index Terms*— Consistency regularization (CR), remote sensing (RS) image, self-training, semantic segmentation, semisupervised learning (SSL).**

I. Introduction

With the development of remote sensing (RS) technology, RS images have significantly improved in quantity and quality. The spatial resolution of a vast amount of data has reached sub-meter levels, resulting in RS images that contain increasingly abundant information [1], [2]. RS semantic segmentation, which entails labeling each pixel of images to extract information, has drawn a lot of interest due to its importance in applications such as ecological conservation and urban planning. Deep-learning techniques [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] are the current prevailing methods for RS semantic segmentation. These methods have demonstrated promising results; however, they rely heavily on fully supervised learning, requiring pixel-level annotations for each RS image.

Annotating RS images is more time-consuming and labor-intensive than annotating natural images. While it takes approximately 0.024 h to annotate a natural image [13], annotating a high-resolution scene image can take up to 1.5 h [14]. Moreover, the annotation of RS images necessitates the involvement of experts with a background in RS because of the complexity and specialized knowledge required. This annotation process incurs high costs, resulting in a scarcity of labeled RS data despite the abundance of RS data. Unfortunately, the potential of these unlabeled data remains largely untapped.

To mitigate these challenges, semi-supervised methods have been developed to leverage labeled and unlabeled RS data. Consistency regularization (CR) [15], [16], [17], [18], [19] and pseudo-labeling-based self-training [20], [21], [22], [23] are commonly used techniques in semi-supervised learning (SSL). CR ensures that consistent outputs are produced when different perturbations of the same data are provided as input. Self-training involves utilizing pseudo-labels produced by a model that has been trained on labeled data and the corresponding images to train the model further. In practice, these two methods are often combined to enhance the performance of semi-supervised RS image semantic segmentation. By leveraging the benefits of both CR and self-training, the model can learn from labeled and unlabeled data, leading to improved segmentation results in RS applications [22], [24], [25]. While these methods show promise, the labeled-to-entire-dataset ratio is often around 1/8 [16], [22], [24], [25].

However, in practical RS applications, the labeled-to-all data ratio could be highly imbalanced, and the current labeled-to-all data ratio is much lower than 1/8, presenting additional challenges. First, semantic segmentation encounters class

Manuscript received 30 November 2023; revised 14 April 2024; accepted 6 May 2024. Date of publication 30 May 2024; date of current version 19 September 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62201550 and in part by the Key Deployment Program of the Chinese Academy of Sciences under Grant KGFZD-145-23-18. *(Corresponding authors: Wanxuan Lu; Hongfeng Yu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3407142

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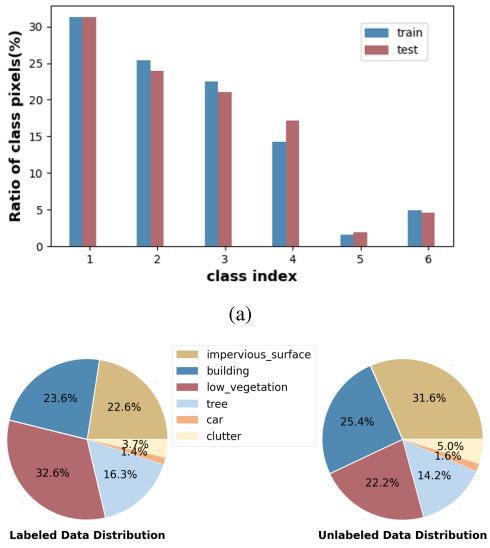


Fig. 1. Imbalanced class distribution within the Potsdam dataset. (a) Pixel class distribution within the Potsdam dataset. Both the training and test sets exhibit an imbalanced class distribution. The 1–6 class indices correspond to impervious surface, building, low vegetation, tree, car, and clutter. (b) Pixel class distribution of the labeled and unlabeled datasets in the 1/32 labeled data ratio (where the number of labeled data is 1/32 of the total training set). We observe that they have different distributions.

imbalance issues due to the varying sizes of objects [26]. Categories with larger objects have more labeled samples, while smaller ones have fewer labeled samples. This imbalance poses difficulties in segmenting smaller, scattered objects compared to larger, contiguous objects. In the Potsdam dataset, the "impervious-surface" category includes roughly 20 times more pixels than the "car" category, as seen in Fig. 1. Consequently, models tend to overfit categories with more samples, while categories with fewer samples are easily neglected.

Second, in SSL, extremely low labeled-to-entire-dataset ratios might result in significant distributional discrepancies between labeled and unlabeled data [27]. It is clear from Fig. 1 that in the 1/32 labeled-to-entire-dataset case, the labeled data in the Potsdam dataset has a different class distribution than the unlabeled set. Such disparities may introduce biases when training models to predict labels for the unlabeled data.

Third, a supervised model with a threshold for filtering out low-confidence pseudo-labels is usually used in SSL to assign pseudo-labels to unlabeled data [16], [28]. However, this approach may mistakenly filter correct pseudo-labels due to difficulties in learning certain classes or limited samples. Additionally, the confidence of pseudo-labels fluctuates during training, resulting in unstable pseudo-label quality. Li et al. [16] modify thresholds according to pseudo-labels for each training image, but they disregard the global class thresholds, which consider the correlation between the same classes across different images.

To tackle the challenges mentioned above, we present a self-training method called dynamic and adaptive self-training (DAST), which integrates CR and addresses the issues of class distribution imbalance and distribution differences between labeled and unlabeled data. DAST employs dynamic pseudo-label sampling (DPS) to alleviate the class distribution imbalance by prioritizing classes with fewer samples. It incorporates distribution matching (DM) and adaptive threshold to mitigate distribution differences by adjusting predictions across augmented images in CR, aligning them with the actual data distribution. Initially, pseudo-labels are assigned to unlabeled data leveraging a supervised model trained with labeled data. To tackle class imbalance, we employ DPS, prioritizing classes with fewer samples to increase the probability of obtaining high-quality pseudo-labels for these classes. This ensures a more balanced data distribution while retraining, which improves the learning and generalization capabilities of the model across diverse classes. We first add DM during retraining, using two distributions: one reflecting the true global distribution obtained from evaluating all trained data, and the other representing the distribution of the current model with a buffer containing pseudo-labels for unlabeled input. We reduce differences between pseudo-labels and actual labels by aligning the model's predictions with the true labels and modifying the supervision targets according to the ratio between these distributions. This enhances the accuracy and generalizability of the model. Additionally, we incorporate adaptive threshold updating (ATU), which modifies thresholds for each class based on the data batches during training. This adaptive approach enables us to update the global class thresholds accordingly. Using more accurate thresholds, we effectively filter out noisy pseudo-labels and obtain high-quality ones, improving overall model performance. This adaptive threshold mechanism enhances the model's prediction reliability and contributes to more accurate segmentation results.

By adopting this self-training method, our objective is to help improve the quality of pseudo-labels, tackle class imbalance, alleviate distributional differences, and enhance the stability of pseudo-label quality. The suggested strategy could deal with the issues posed by significant disparities in the labeled-to-unlabeled data ratio in RS semi-supervised segmentation.

The following concisely describes the primary contributions of our work.

1) We present a novel SSL framework, DAST, specifically designed to address the challenge of significantly disparate labeled-to-unlabeled ratios in RS semi-supervised segmentation.
2) Our framework incorporates DPS based on class distribution, effectively tackling the issue of class imbalance. By assigning pseudo-labels to unlabeled data in a manner that accounts for the distribution of different classes, we ensure that classes with fewer samples are given higher priority, leading to improved segmentation results.
3) We have proposed a prediction adjusting method in CR which incorporates the DM module and adaptive threshold module that enables our models to learn a more accurate representation of the underlying data distribution. By aligning the model's distribution with the actual global distribution of the data and filtering out noisy targets, we improve the model's generalizing ability and make more reliable predictions.

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4) We have demonstrated the efficacy of our DAST method through extensive experiments on the Potsdam and iSAID datasets. The results validate the improvements achieved in RS semi-supervised segmentation tasks, further highlighting the significance of our contributions.

II. Related Works

We will quickly go over a few pertinent studies on semantic segmentation in RS images in this section. Then, we will discuss some semi-supervised semantic segmentation research. Finally, we will concentrate on the semantic segmentation of semi-supervised RS. Semantic segmentation in RS divides RS images into regions with semantic information, commonly used for monitoring and researching natural and man-made environments. Semi-supervised semantic segmentation trains neural networks for image segmentation by utilizing labeled and unlabeled samples, to improve segmentation accuracy and robustness. Applying semi-supervised semantic segmentation to RS images, so-called semi-supervised RS semantic segmentation enables better extraction of RS of semantic information to support more accurate RS applications, such as resource management, disaster response, and urban planning.

A. Semantic Segmentation of RS Images

Images from RS are not like those from natural scenes; they often contain small structures or objects with smaller sizes or finer shapes, such as cars and building edges. The shadows generated by buildings can also negatively affect RS images. Therefore, achieving precise RS image segmentation remains a challenging task. With transformers' popularity, RS image segmentation has also started exploring transformer frameworks and attention mechanisms. Compared to CNNs, transformers perform better in capturing global information [29], [30], [31], making them suitable for experiments in RS semantic segmentation tasks [10], [32], [33].

Wang et al. [10] suggest a new architecture for deep neural networks called UNetFormer, designed for efficient and accurate pixel-level semantic segmentation of urban scenes. UNetFormer combines a CNN encoder with a Transformer decoder, where the decoder employs global–local transformer blocks (GLTBs) to capture global and local contextual information. It also introduces feature refinement heads (FRHs) to improve the quality of feature maps further. Sun et al. [11] recommend a continuous pooling attention network (SPANet), which combines multiscale feature extraction and self-attention mechanisms, enabling the extraction of richer features while maintaining the continuity and stability of the network. Wang et al. [32] suggest a brand-new transfer domain adaptive technique for images from RS (TDARS). To improve the relationship between the given source domain and the target domain and to gain knowledge from both major and minor reliable classes, they introduce an intermediate domain, which increases the possibility of important information transfer. He et al. [12] propose a novel framework called ST-UNet for RS image semantic segmentation, integrating the Swin Transformer's [34] potent global modeling capability. Compared to most existing CNN-based methods, ST-UNet integrates the Swin Transformer into a traditional CNN-based U-Net network to capture global contextual information. ST-UNet adopts a new dual-encoder framework, leveraging parallel computations between the Swin Transformer and CNN. Due to the great performance in capturing global contextual information, we base our model on the Swin Transformer [34].

Although the emergence of Transformers has extensively improved the efficacy of RS image semantic segmentation [35], the above-mentioned methods are all based on labeled images for training. However, obtaining labeled images for RS is challenging and expensive. It is noted that there are numerous unlabeled RS images available, constantly being generated. Therefore, leveraging many unlabeled images to enhance the performance of RS semantic segmentation can be considered.

B. Semi-Supervised Semantic Segmentation

The research community has focused a lot of attention on semi-supervised semantic segmentation to increase the practicality of deep-learning techniques [17], [21], [36], [37], [38], [39], [40], [41], [42], [43], [44] because labeled data acquisition is expensive. We will discuss several cutting-edge methods for semi-supervised semantic segmentation below.

Some methods focus on self-training. Ke et al. [38] propose a three-step self-training method. It first trains the segmentation network using labeled data only and generates coarse pseudo-masks for all images, then uses a multitask model to reduce the uncertainty in pseudo-masks and leverages abundant statistical knowledge within the data to maintain consistency. At last, they train the segmentation model using higher-quality pseudo-mask information. Yang et al. [21] construct a strong self-training baseline (referred to as ST) by applying strong data augmentations (SDAs) to unlabeled images to distinguish similar predictions made by students and teachers and reduce overfitting to label noise. They propose ST++ that saves multiple model checkpoints from the first-stage supervised training and uses their prediction discrepancies as a reliability measure.

Some methods focus on CR and high-quality pseudo-label selection. Liu et al. [39] introduce auxiliary teacher networks to improve the segmentation accuracy on unlabeled images. It replaces the original mean squared error (mse) loss function with a stricter confidence-CE loss function to enhance training convergence and accuracy. Wang et al. [40] introduce an unbiased subclass regularization network (USRN) that mitigates class imbalance issues by constructing balanced subclass distributions. Sohn et al. [41] produce pseudo-labels for weakly augmented unlabeled images based on model predictions; pseudo-labels are only kept for images with high-confidence predictions. It inspires many subsequent semi-supervised methods. Wang et al. [42] adaptively adjust the confidence threshold to make more efficient use of unlabeled data. It introduces adaptive class fairness regularization penalties to encourage model diversity in early training stages. Chen et al. [43] propose a confidence-based truncated Gaussian weighting of samples, which is comparable to the confidence threshold but less strict. It further introduces a unified

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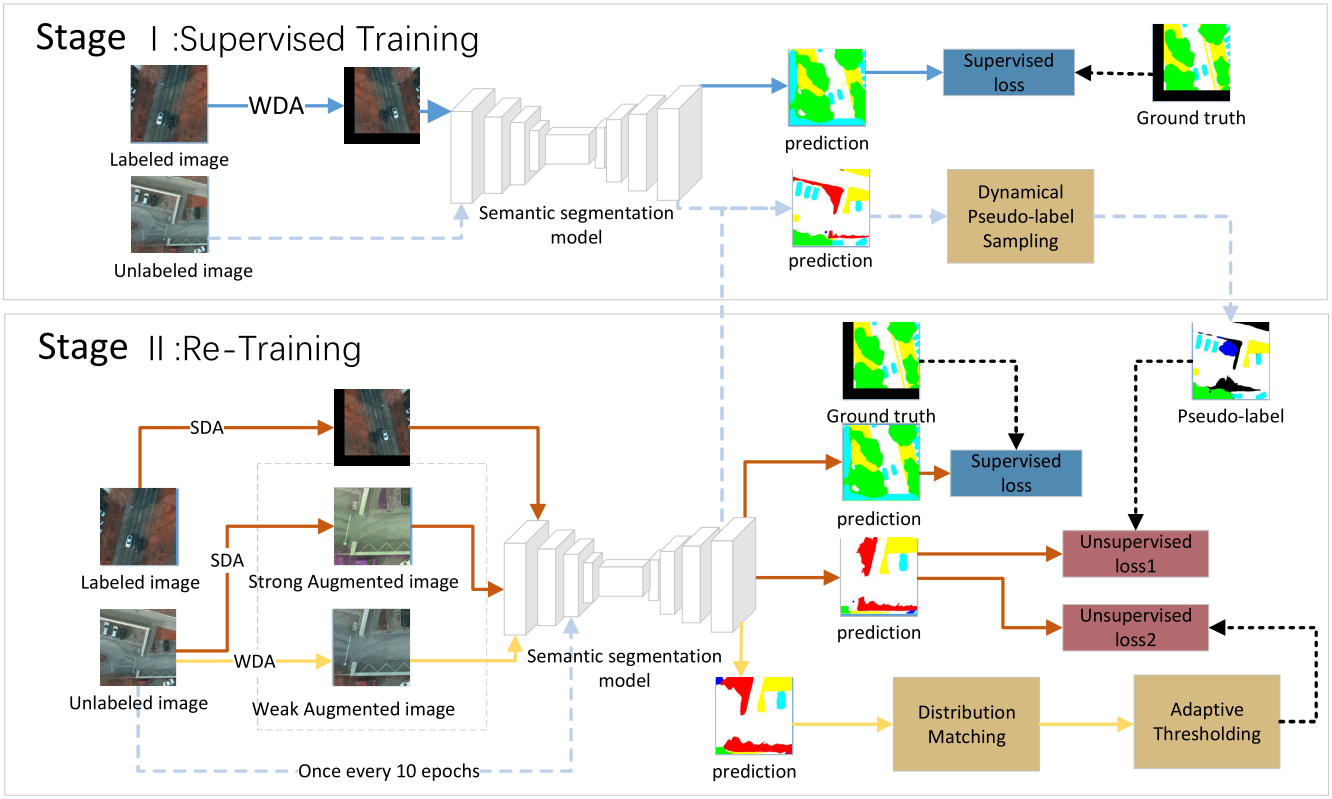


Fig. 2. Overview of the DAST framework. We first train a supervised model using the labeled data and leverage it to predict the pseudo labels of unlabeled images. Then we sample the pseudo-labels by the dynamical pseudo-label sampling module. For retraining, we apply weak and strong augmentation in unlabeled images separately. The prediction of the weak augmented image will be through the DM module and filtered by the adaptive thresholding module to get the target signal, which is used to supervise the prediction of the strong augmented image by **unsupervised loss2**. We will also use the pseudo-labels to supervise the prediction of the strong augmented image by **unsupervised loss1**. We update the pseudo-labels every 10 epochs by the dynamical pseudo-label sampling module as shown in the process outlined in the figure's white dashed line. WDA and SDA refer to "weak data augmentation" and "SDA" respectively.

alignment method to enhance the utilization of weakly learned classes.

The success of semi-supervised semantic segmentation in natural scenes pushes researchers to introduce SSL to the RS region.

C. Semi-Supervised Semantic Segmentation in RS Images

In recent years, researchers have started exploring semi-supervised techniques for semantic segmentation in RS. Here are some cutting-edge techniques in semi-supervised RS image semantic segmentation.

Li et al. [16] propose a strategy based on a generative adversarial network (GAN) framework [45]. By evaluating the confidence of each pixel's prediction using pixel-level training labels, the method optimizes the discriminator and uses the results to reweight the loss of the unlabeled data. He et al. [15] introduce an SSL framework based on a hybrid perturbation paradigm. It considers perturbations at both the data and model levels, integrating semantic boundary information into the data-level perturbation to generate more meaningful mixed images.

Lu et al. [22] propose an SSL framework based on linear sampling and self-training, called LSST. They design SDAs suitable for RS images and adaptively assign thresholds for different classes to provide high-quality pseudo-labels for retraining. Wang et al. [25] propose a method that generates pseudo-labels using a random pasting approach, where some real labels are pasted into predicted segmentation images. Then they use the aforementioned pseudo-labels to supervise the predictions of training images transformed by SDA with threshold adjustments.

These techniques aim to leverage the benefits of SSL to enhance the performance of RS image semantic segmentation, utilizing the abundance of unlabeled data.

III. Methods

To address the issues of class distribution imbalance and distribution differences between labeled and unlabeled data, we introduce a self-training method called DAST, which integrates CR. Fig. 2 presents three key components: DPS based on class distribution, DM, and ATU. We will give a general overview of the framework and go into more detail about each module in the following sections.

A. Problem Formulation and Overall Structure

Semi-supervised semantic segmentation is defined as having two sets of data: one with labeled data, X_L, and corresponding labels, Y_L, and another with unlabeled data, X_U. In reality, labeled data is often much scarcer than unlabeled data. The main focus of semi-supervised semantic segmentation is how to leverage the finite amount of labeled data and the abundance

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of unlabeled data to improve the segmentation performance of the model. Standard methods include CR and self-training.

CR aims to ensure that the same output is obtained when different perturbations of the same data are provided as input. Strong and weak augmentation are used to create augmented samples for unlabeled data, separately denoted as X_S and X_W. These samples are input into the model, resulting in predicted outputs Y_S and Y_W

$$Y_S^i, Y_W^i = \widehat{P}\left(X_S^i; \Theta\right), \quad \widehat{P}\left(X_W^i; \Theta\right) \tag{1}$$

where $\widehat{P}(X_S^i; \Theta)$ means use the model of parameter Θ to predict the segmentation of X_S^i. And X_S^i represents the strongly augmented image of the ith image X^i. The cross-entropy loss between the predictions is calculated as follows and used for model training:

$$\mathcal{L}_{\text{ce}}\left(Y_S^i, Y_W^i\right) = -\sum_{c \in C} Y_W^i[c] \cdot \log\left(Y_S^i[c]\right). \tag{2}$$

Self-training typically involves supervised training of the model using labeled data initially. Then, unlabeled data will be assigned pseudo-labels using the trained model. Finally, the model is supervised by the original labels and the pseudo-labels.

Our method follows the self-training paradigm and consists of two stages. First, we conduct supervised training using labeled data, resulting in multiple checkpoints. We select the best one among these checkpoints based on the mean intersection over union (mIoU) metric. Then, we produce pseudo-labels for the unlabeled data leveraging the best checkpoint obtained in the first step. Pseudo-labels are predicted for the unlabeled images, and we sample them according to our selection criteria. We employ the DPS module to obtain the final set of pseudo-labels. The second stage involves retraining the model with the original labels from the labeled data and the pseudo-labels generated in the first stage. In this stage, there are two branches. In one branch, the predictions of strongly augmented images are supervised by pseudo-labels, similar to the first stage. In the other branch, the predictions of strongly augmented images are supervised by new target labels. These target labels are generated from weakly augmented samples, which are obtained through the DM module and filtering based on adaptive thresholds. The adaptive *threshold updating* module updates the thresholds used for filtering.

B. Dynamical Pseudo-Label Sampling

The quality of pseudo-labels determines the effectiveness of self-training methods, making the selection method for pseudo-labels crucial. Typically, a threshold needs to be set to filter out low-confidence noise labels and improve the quality of pseudo-labels. However, as the model training progresses, confidence levels may vary significantly, rendering a fixed threshold unsuitable. Existing methods [22], [24], [25] not only dynamically adjust thresholds based on the training process but also assign different thresholds to different classes. However, they overlook the issues of class imbalance and the distribution of information embedded in the historical data.

LSST [22] addresses these concerns by assigning different thresholds to different classes based on the current confidence of pseudo-labels and designing a fixed ratio to determine the proportion of pixels to be sampled for each class. It takes into account the varying difficulty levels of different classes but fails to consider the issue of changing distributions due to the fixed ratio. Wang et al. [24] do not explicitly set thresholds or sample pseudo-labels but instead average the current and previous pseudo-labels during training to obtain adaptive pseudo-labels. While this method considers the need for pseudo-label adaptation during model training, it does not address class imbalance or distribution issues. Ranpaste [25] tackles the problem of overfitting by adaptively setting thresholds for pseudo-labels based on the confidence of class predictions across the entire image. It removes high-confidence pixels to prevent overfitting and low-confidence pixels to improve the precision of pseudo-labels. However, it does not assign different thresholds to different classes or consider distribution issues.

To address the issue of class imbalance in RS data, we sample pseudo-labels dynamically. Initially, we assume the data of different classes follow a uniform distribution. As the training progresses, we collect statistics on the class distribution from the pseudo-labels of the previous sampled round. Based on this distribution, we calculate the sampling rate for each class, which is inversely proportional to the number of samples, indicating classes with higher sample quantities will have lower sampling rates. Distribution of pseudo-labels is defined as $[N_0, N_1, \ldots, N_C]$, $N_0 < N_1 < \cdots < N_C$. The formula to calculate the sampling rate p_c is derived from the distribution

$$p_c = \left(\frac{N_{C-c}}{N_0}\right)^{\frac{1}{\alpha}} \tag{3}$$

where α is a hyperparameter to adjust the sampling curve.

DPS is an under-sampling method, which primarily focuses on reducing the dominance of head classes, alleviating model overfitting toward abundant classes, thereby enhancing the model's performance by increasing the prediction probabilities for other classes. Moreover, since the confidence of minority classes is often lower, our under-sampling method tends to retain a significant portion of pseudo-labels for these challenging classes while discarding some pseudo-labels for simpler majority classes, effectively improving the quality of pseudo-labels. Two other common methods for addressing class imbalance are over-sampling and re-weighting. Over-sampling aims to increase the sample size of tail classes, but it risks overfitting due to limited variability. Re-weighting assigns different weights to samples in the loss function, often favoring tail classes, but requires hyperparameter fine-tuning and may lack generalization, especially in semi-supervised tasks with unknown data distributions. Consequently, we introduce DPS to address class imbalance issues.

To elaborate, we first leverage the model to predict labels for the unlabeled data. Then, we apply a confidence threshold to filter the predictions and obtain segmentation maps. We consider each connected component of pixels with the same class label in a segmentation map as a region (such as the colorful region in Fig. 3), defined as the minimum sampling units. This approach significantly reduces the computational burden. We sample regions from all images based on corresponding

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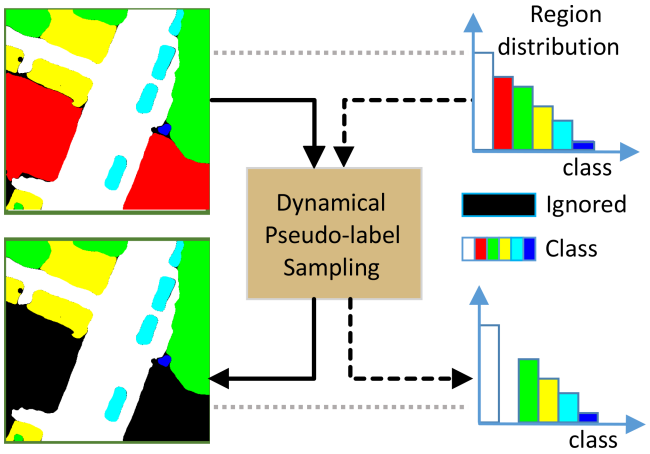


Fig. 3. Illustration of dynamical pseudo-label sampling. We use a trained model to predict unlabeled images and cluster the adjacent pixels of the same class to a region. Then, we sample the region according to the region distribution and confidence. After sampling, we can see that the red class, which has more samples, has been neglected. The upper image shows small regions that were initially overlooked (in black), while the lower image displays both the initially ignored small regions and the regions that were filtered after sampling (also in black).

class sampling rates, prioritizing higher-confidence regions. The confidence of a region is defined as the average confidence of all pixels within that region.

During retraining, we perform pseudo-label sampling every 10 epochs to update the distribution of pseudo-labels and obtain more accurate pseudo-labels, which enhances the model performance. When the sampling frequency is too high, the model has not learned enough before generating pseudo-labels, leading to noise that cannot be eliminated through training. Conversely, when the sampling frequency is too low, the model keeps learning the same knowledge without performance improvement. Consequently, we choose 10 epochs as the sampling frequency.

C. Distribution Matching

There is usually a distribution mismatch between labeled and unlabeled data because of the small amount of labeled data relative to unlabeled data. When using a model trained on labeled data to predict unlabeled data, the model tends to overfit the distribution of the labeled data, resulting in less accurate predictions for the unlabeled data. Therefore, we incorporate the unlabeled data for distribution adjustment when calculating the loss of CR.

Specifically, we introduce two variables: $D_{\text{data}} \in \mathcal{R}^C$ and $D_{\text{model}} \in \mathcal{R}^C$, separately denoting the distribution of data and distribution of the model. Every element in D_{data} and D_{model} is initialized as $(1/C)$. we denote $\text{Buf}_{\text{model}} \in \mathcal{R}^{C\times M}$ as a buffer to store the predictions of M unlabeled data samples. After each batch training, we use the model predictions Y_W of weakly augmented data to update these two variables. After updating the predictions of weakly augmented images, D_{data} can signify the extent of training data deviation from the average distribution, effectively portraying the distribution of training data. The capacity of $\text{Buf}_{\text{model}}$ should be proper, ensuring it is neither too diminutive to represent the learned distribution nor excessively large to drive the model's distribution toward the data distribution, thereby forfeiting its corrective significance. The updating formulas are as follows:

$$D^t_{\text{data}} = D^{t-1}_{\text{data}} * \text{decay} + Y^t_W * (1 - \text{decay}) \tag{4}$$

$$\text{Buf}^t_{\text{model}} = \text{popandpush}\left(\text{Buf}^{t-1}_{\text{model}}, Y^t_W\right) \tag{5}$$

where the former Y_W represents hard labels and the latter Y_W represents soft labels. The functionpopandpush() means that we first pop $\text{Buf}_{\text{model}}$ and then push Y_W into $\text{Buf}_{\text{model}}$. Then D_{model} is calculated by averaging $\text{Buf}_{\text{model}}$

$$D^t_{\text{model}} = \text{Mean}\left(\text{Buf}^t_{\text{model}}\right). \tag{6}$$

We then adjust the distribution of Y_W using D_{data} as the target distribution and D_{model} as the model distribution. The final label $\widetilde{Y}_W$ is calculated as follows:

$$\widetilde{Y}_W = \text{argmax}\left(Y_W \odot \frac{D_{\text{data}}}{D_{\text{model}}}\right). \tag{7}$$

It is noted that the multiplication and division operations are all element-wise operations. The CR loss is computed as follows:

$$\mathcal{L}_{\text{CR}} = \text{CE}\left(Y_S, \widetilde{Y}_W\right) \tag{8}$$

where Y_S represents the prediction of strong augmented data.

By DM module, we mitigate the bias caused by the distribution mismatch and encourage the model to predict the unlabeled images more accurately.

Algorithm 1 Retraining of Self-Training

Input:
$\mathcal{U} = (u^b, y^b_u; b \in (1, \cdot, B))$: data of unlabeled images and the corresponding pseudo target

Output:
Θ_{best}: optimal model parameters;

for $t = 1$ to T **do**
  for $b = 1$ to B **do**
    $u^b_s = StrongAugment(u^b)$ //Apply strong augmentation to u^b;
    $u^b_w = WeakAugment(u^b)$ //Apply weak augmentation to u^b;
    $q_b = P_{model}(y|u^b_w; \theta)$;
    $Q_b = P_{model}(y|u^b_s; \theta)$;
    $\hat{q}_b = Distribution_match(q_b)$//adjust the pseudo target;
    Filter $\hat{q}_b$ according to adaptive thresholding;
    Calculate the unsupervised loss:$\mathcal{L}_U = CE(Q_b, \hat{q}_b)$;
    Calculate the supervised loss:$\mathcal{L}_S = CE(Q_b, y^b_u)$;
    Integrate all losses: $\mathcal{L} = \mathcal{L}_U + \mathcal{L}_S$;
    Back propagating the total loss and update Θ
  end for
  if $t \mod 10 = 0$ **then**
    Mark the current model checkpoint and use it to predict pseudo labels of unlabeled images;
    save the best checkpoint Θ_{best}
  end if
end for
return Θ_{best}

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D. Adaptive Thresholding

Since different classes may exhibit varying levels of segmentation difficulty, it is not reasonable to use a fixed threshold to filter labels for all classes. Therefore, we assign different thresholds to different classes and dynamically adjust them during retraining. We set an initial global uniform threshold, $T = (1/C)$, where C is the number of classes. During each batch of training, we update the global threshold as follows:

$$T = T * \text{decay} + (1 - \text{decay}) * T_B \tag{9}$$

where T_B represents the average confidence of all pixel segmentations in the current batch.

To adjust the confidence thresholds for each class, we use the varying data distribution Y_W with retraining. Specifically, we calculate T_c that represents the adjusted threshold for the class c like

$$T_c = T * \frac{Y_{W_c}}{\max(Y_W)}. \tag{10}$$

When generating the target labels in the CR process, we apply T_c to filter the pixels of different classes. This approach raises the quality of the target labels and diminishes the learning of incorrect labels. By dynamically adjusting the confidence thresholds based on the data distribution, we adaptively handle the segmentation difficulty of different classes and optimize the learning process for improved performance.

IV. Experiments

A. Datasets

1) POTSDAM [46]: The Potsdam dataset is commonly utilized for RS image segmentation tasks. It consists of high-resolution aerial images from Potsdam, Germany. There are 38 aerial images in the dataset, each with a resolution of 5 cm/pixel and a size of 6000 × 6000 pixels. These images cover various scenes in and around Potsdam, including urban areas, rivers, forests, grasslands, and farmland. Each pixel in the Potsdam dataset is labeled into one of six classes: Impervious Surface, Building, Low Vegetation, Tree, Car, and Clutter. The dataset provides corresponding labels for each pixel, which can be used for training and evaluating the performance of RS image segmentation algorithms. The Potsdam dataset has been widely used in research and competitions, such as the ISPRS 2-D Semantic Labeling Contest. Its large scale and diversity make it an ideal benchmark dataset for evaluating RS image segmentation algorithms. We used 24 images for training and 14 for validation and testing. To facilitate training, we divide the images into nonoverlapping patches of size 512 × 512 pixels. The training set consists of 3456 patches, while the testing set consists of 2016 patches. Typically, only a fraction of the training set (1/32) is used as labeled data, which amounts to 108 patches. Unless otherwise specified, the remaining patches (3348) are treated as unlabeled data.

2) ISAID [47]: The iSAID dataset comprises 2806 RS images captured by GF-1 and GF-2 satellites. This dataset is widely employed in semantic segmentation research and applications. It includes 1411 images in the training set, 458 in the validation set, and 937 in the test set(unavailable) with a wide range of sizes. There are 655 451 instances of 15 object categories and one background class included in the dataset annotations. The following categories are listed in alphabetical order: Background, Baseball Diamond, Basketball Court, Bridge, Ground Track Field, Harbor, Helicopter, Large Vehicle, Plane, Roundabout, Ship, Small Vehicle, Soccer Ball Field, Storage Tank, Swimming Pool, and Tennis Court. Like the Potsdam dataset, we divide the training and validation sets into nonoverlapping patches of 512 × 512 pixels.

B. Metrics

For pixel-level semantic segmentation, we employ the MIoU [48], which is the average value of the intersection over the union (IoU) of all classes. The IoU is a metric that calculates the overlap between the predicted and actual (ground truth) segmentation masks, divided by the union of these masks. IoU values fall within the range of 0–1 and provide a robust measure for evaluating segmentation results, even in scenarios with an imbalanced distribution of classes within the datasets. We also use overall accuracy (OA) and F1 to analyze the experiment result. The definitions are as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{FN} + \text{FP} + \text{TP}} \tag{11}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{N} \tag{12}$$

$$\text{F1} = (1 + \beta)\frac{\text{precision} \cdot \text{recall}}{\beta^2(\text{precision} + \text{recall})}, \quad \beta = 1 \tag{13}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{15}$$

where true positives (TP) represent the pixels that are correctly classified, false positives (FP) represent pixels that are incorrectly classified, false negatives (FN) are pixels that are incorrectly rejected, and true negatives (TN) are pixels that are correctly rejected. Then, MIOU is calculated as follows:

$$\text{MIoU} = \frac{\sum_{i=1}^{C} \text{IoU}_i}{C} \tag{16}$$

where IoU_i means the IoU of the ith class and C is the number of pixel classes.

C. Implementation Details

We use PyTorch for implementing the experimental code. The semi-supervised model utilizes the Swin transformer [34] as the backbone and Uperhead [49] as the segmentation head. Additionally, we employ FCNhead [50] as an auxiliary segmentation head. The labeled-to-entire-dataset ratios for the Potsdam and iSAID datasets are at least 1/32.

During training, both the first stage of supervised training and the second stage of retraining employ the same set of hyperparameters. All datasets are trained using four NVIDIA RTX 3090 GPUs, with a batch size of 8. We use the AdamW optimizer with a learning rate of 0.00006 and a weight decay of 0.01. No weight decay is applied to position embedding

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TABLE I
PERFORMANCE COMPARISON ON **POTSDAM** *Valset With Clutter* WITH THE LABELED DATA RATIO OF **1/32**.WE REPORT EACH CATEGORY'S PERFORMANCE ON MIOU, OA, MF1, AND F1 SCORES

Ratio	Method	mIoU↑	OA↑	mF1↑	**F1** per category					
					Imp. Surf.	**Building**	**Low Veg.**	**Tree**	**Car**	**Clutter**
1-32	**Baseline**	66.37	84.40	77.36	86.73	90.71	81.77	83.09	87.36	34.51
	LSST [22]	60.82	80.57	74.15	83.04	87.34	77.21	78.39	76.87	42.04
	ST++ [21]	64.89	83.83	75.37	86.78	89.39	81.64	82.93	88.18	23.32
	ClassHyPer [15]	66.58	83.62	78.23	86.40	90.62	80.42	82.69	85.71	43.55
	DAST(Ours)	**71.83**	**87.07**	**82.05**	**90.02**	**94.10**	**83.08**	**85.23**	**89.91**	**49.94**

and layer normalization in the backbone. For simplicity, we did not update the learning rate during training. Then, we train the semi-supervised segmentation model using fractions of labeled images, which are 1/4, 1/8, 1/16, and 1/32, along with the remaining unlabeled images.

In both the first and second stages, we train for 80 epochs. In the first stage, we record a checkpoint every 10 epochs. We utilize the best checkpoint from the first stage to assign pseudo-labels to the unlabeled data for the initial training in the second stage. In the second stage, we update the pseudo-labels every 10 epochs using the current model and record the corresponding checkpoint. We evaluate the models using the best checkpoint from the second stage to ensure fair comparison.

D. Experimental Results of the POTSDAM Dataset

This section will show the result of the POTSDAM dataset with different data ratios. First, we will show the result with the labeled data ratio of 1/32. Then, according to other related articles, we also show the results of commonly labeled data ratios, 1/16, 1/8, and 1/4.

1) Results of 1/32 Labeled Data Ratio: Table I presents a comprehensive comparative analysis within our method and several other semi-supervised approaches on the Potsdam test dataset, focusing on a 1/32 labeled data ratio. The table clearly illustrates that our approach significantly improves various metrics. Specifically, we observe substantial enhancements in mIoU, OA, and mean F1 scores, with improvements of 5.46%, 2.67%, and 4.69%, respectively, when compared to the baseline. Furthermore, it is worth noting that ClassHyper [15] achieves higher mIoU and mean F1 scores than our baseline by 0.21% and 0.87%, primarily attributable to its superior performance in the clutter class. Our improvements across various classes, including Impervious Surface, Building, Low Vegetation, Tree, Car, and Clutter, amount to 3.29%, 3.39%, 1.31%, 2.14%, 2.55%, and a remarkable 15.43%, respectively, when contrasted with the baseline. The most significant enhancement in the clutter class demonstrates our method's proficiency in capturing complex object features. It is noteworthy that ST++ [21] surpasses ClassHyper [15] in F1 scores per category, except for building and clutter, with similar F1 scores in Low Vegetation and Car. Nevertheless, our method outperforms ST++ [21] by 6.68% in mean F1, notably achieving a substantial 26.62% performance boost in the clutter class. This highlights that ST++ [21] struggles with an abundance of false image-level pseudo-labels, while our method effectively mitigates this issue.

2) Results of Common Labeled Data Ratios: Moving on to the common labeled data ratios results, Table II presents the comparative analysis at 1/16, 1/8, and 1/4 labeled data ratios. In scenarios with a 1/16 labeled data ratio, our method enhances mIoU, OA, and mean F1 by 4.77%, 2.20%, and 3.78% compared to the baseline. At a 1/8 labeled data ratio, our method achieves improvements of 4.50% in mIoU, 2.18% in OA, and 3.26% in mean F1 compared to the baseline. Even with a 1/4 labeled data ratio, our approach still delivers enhancements of 2.10% in mIoU, 0.92% in OA, and 1.49% in mean F1 compared to the baseline. This analysis underscores that our method excels in scenarios with limited labeled data. The most substantial mIoU improvement, 5.46%, occurs at the 1/32 labeled data ratio, followed by a 4.77% improvement at the 1/16 labeled data ratio. Furthermore, it is worth mentioning that, except for the 1/4 labeled data scenario, all other advanced methods fail to outperform the baseline. This reaffirms the effectiveness of our strong baseline, which involves training the Swin-Tiny transformer using corresponding labeled data at each data ratio, highlighting the robust performance of the transformer. Notably, when the same baseline is applied to our method and ST++ [21], ST++ [21] yields worse results than the baseline, decreasing mIoU by 1.48%, 0.99%, 2.22%, and 3.94% in 1/32, 1/16, 1/8, and 1/4 labeled data scenarios compared to the baseline. This underscores that our method is more suitable for SSL with Swin transformers than ST++ [21].

3) Supplementary Experiments: Additionally, we conduct a detailed comparison with existing prominent SSL methods. To directly compare with Li et al. [16], we calculate our method's mean IoU and mean F1 without considering the clutter class in 1/8 and 1/4 labeled data scenarios. In the 1/8 labeled data scenario, our approach achieves mean IoU and mean F1 scores of 80.53% and 89.07%, surpassing Li et al. [16] by 3.61% and 2.25%, respectively. In the 1/4 labeled data scenario, our method obtains mean IoU and mean F1 scores of 80.77% and 89.22%, outperforming Li et al. [16] by 3.38% and 2.14%, respectively. Moreover, it is important to note that the Potsdam dataset used by Li et al. [16] is

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TABLE II
PERFORMANCE COMPARISON ON **POTSDAM** WITH THE COMMON RATIO (IT IS NOTICED THAT LI ET AL. [16] IGNORE THE CLUTTER AND *italic-underline* IS THE MEAN METRIC WITHOUT THE CLUTTER)

Ratio	Method	mIoU↑	OA↑	mF1↑	F1 per category					
					Imp. Surf.	Building	Low Veg.	Tree	Car	Clutter
1-16	**Baseline**	67.75	85.28	79.03	88.21	92.00	81.38	83.36	85.03	44.20
	LSST [22]	62.16	81.97	75.00	85.03	89.38	77.27	80.45	76.39	41.46
	ST++ [21]	66.76	85.18	77.24	88.75	92.83	80.74	82.59	87.46	31.09
	ClassHyPer [15]	68.90	85.48	80.05	87.98	93.07	81.93	82.65	86.63	48.03
	DAST(Ours)	**72.52**	**87.48**	**82.81**	**90.47**	**93.96**	**85.63**	**85.26**	**89.13**	**54.46**
1-8	**Baseline**	69.32	85.81	80.53	88.38	93.28	81.36	82.81	86.24	51.09
	LSST [22]	64.00	82.35	76.85	84.14	91.21	77.48	80.41	77.77	50.08
	ST++ [21]	67.10	83.96	78.24	86.06	90.94	79.06	84.61	88.98	39.77
	ClassHyPer [15]	69.41	85.93	80.27	88.46	93.06	82.36	83.43	87.75	46.55
	Li et al. [16]	*76.92*	87.07	*86.82*	89.18	92.59	82.32	83.39	86.63	-
	DAST(Ours)	**73.82**	**87.99**	**83.79**	**90.69**	**95.04**	**84.02**	**85.09**	**90.49**	**57.43**
1-4	**Baseline**	71.93	87.15	82.43	90.10	94.50	83.00	83.64	88.59	54.74
	LSST [22]	64.83	83.24	77.52	84.65	90.63	79.34	83.04	76.19	51.25
	ST++ [21]	67.99	85.55	78.18	87.76	93.62	82.42	84.25	88.57	32.45
	ClassHyPer [15]	69.89	86.26	80.65	88.21	93.22	82.83	84.73	87.60	47.30
	Li et al. [16]	*77.39*	87.26	*87.08*	89.67	93.83	82.75	82.22	86.92	-
	DAST(Ours)	**74.03**	**88.07**	**83.92**	**90.85**	**95.07**	**83.91**	**85.60**	**90.66**	**57.45**

acquired by cropping a large image into smaller segments with overlap. In contrast, our Potsdam dataset is cropped without overlap. Consequently, our dataset contains fewer images compared to Li et al. [16], further underscoring our approach's data efficiency. Our method demonstrates its robustness and efficiency, particularly in scenarios with limited labeled data.

In Fig. 4, our method consistently outperforms other approaches across various labeled data ratios. Notably, our approach exhibits the smallest slope from 1/32 to 1/4, indicating its ability to achieve high data utilization with fewer data instances while maintaining competitive performance.

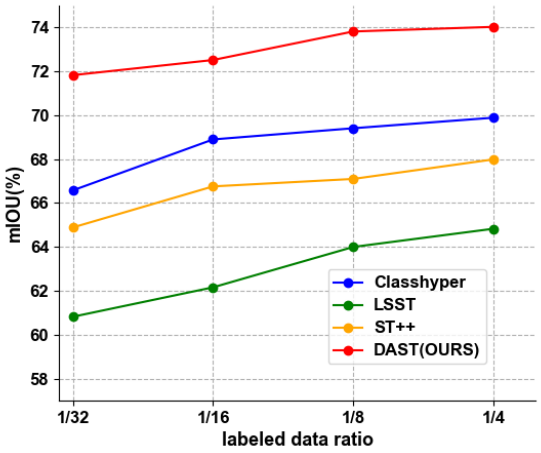


Fig. 4. mIOU curve on the Potsdam dataset.

E. Experimental Results of the iSAID Dataset

This section presents the results obtained on the iSAID dataset, covering various data ratios. Similar to our presentation of the results for the Potsdam dataset, we will delve into the performance outcomes for labeled data ratios of 1/32, 1/16, 1/8, and 1/4.

1) Results of 1/32 Labeled Data Ratio: Tables III and IV present a comparative analysis between our method and several other semi-supervised techniques on the iSAID test dataset. The supervised baseline demonstrates a notable decline in mIoU, plummeting from 59.56% to 44.95% as the ratio of labeled images decreases from 1/4 to 1/32. This stark drop underscores the pivotal role of labeled data quantity in model performance, a challenge that SSL aims to mitigate.

Our framework substantially improves over the 1/32 labeled baselines, achieving a 3.19% increase in mIoU scores. Particularly noteworthy is the significant enhancement in the Roundabout category, where our method records a remarkable 16.03% improvement.

2) Results of Common Labeled Data Ratios: Regarding 1/16 labeled data, our approach elevates the mean IoU by 1.77% compared to the baseline. Notably, the mean IoU of ST++ [21] surpasses our baseline by 1.15%, with only a marginal 0.72% difference in favor of our method. Moreover, our method outperforms all other advanced techniques, which fall short of even the baseline's performance. A standout

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TABLE III

PERFORMANCE COMPARISON ON **iSAID** *valset* WITH THE LABELED DATA RATIO OF **1/32**. WE REPORT PERFORMANCE ON MIOU AND IOU FOR EACH CATEGORY

Ratio	Method	mIoU↑	IOU per category														
			BG	Ship	ST	BD	TC	BC	GT	Bridge	LV	SV	Heli.	SP	Round.	SBF	Plane	Harbor
1-32	**Baseline**	44.95	98.22	56.69	58.63	32.20	82.75	47.67	24.82	5.32	54.18	**44.05**	8.74	34.40	11.95	28.90	76.74	53.88
	LSST [22]	24.73	73.10	0.95	26.56	33.25	35.17	15.33	16.40	4.30	33.61	20.05	5.60	24.94	3.39	36.19	45.42	21.41
	ClassHyPer [15]	39.03	98.26	38.78	26.17	25.16	**86.80**	23.46	22.93	7.57	56.40	40.01	4.99	**36.76**	10.48	31.26	68.31	47.10
	ST++ [21]	47.67	**98.55**	54.22	59.07	**50.05**	80.69	49.38	**30.26**	7.85	**58.35**	37.32	0.00	27.31	**30.94**	**53.75**	76.87	48.08
	DAST(Ours)	**48.14**	98.46	**59.65**	**62.98**	48.07	85.52	**54.92**	10.90	**13.82**	57.36	41.98	**16.39**	36.28	27.98	22.56	**78.99**	**54.39**

TABLE IV

PERFORMANCE COMPARISON ON **iSAID** *valset* WITH THE COMMON RATIO. WE REPORT PERFORMANCE ON MIOU AND IOU FOR EACH CATEGORY

Ratio	Method	mIoU ↑	IOU per category														
			BG	Ship	ST	BD	TC	BC	GT	Bridge	LV	SV	Heli.	SP	Round.	SBF	Plane	Harbor
1-16	**Baseline**	52.21	98.70	58.31	61.96	54.09	85.76	49.36	26.57	15.75	60.83	44.07	**24.53**	49.57	8.61	62.88	79.09	55.31
	LSST [22]	32.56	95.29	4.37	37.71	41.74	61.05	19.53	24.52	4.45	39.97	22.22	0.00	23.11	17.55	40.26	52.70	36.43
	ClassHyPer [15]	40.84	98.33	49.29	38.93	33.99	79.83	44.28	20.54	4.64	54.92	44.12	0.37	30.17	6.39	29.17	68.54	49.88
	ST++ [21]	53.26	98.68	**60.20**	**62.94**	**64.96**	**88.31**	40.70	35.58	14.43	**61.06**	**46.85**	22.52	**53.33**	**27.12**	39.78	79.38	**56.38**
	DAST(Ours)	**53.98**	**98.72**	55.89	60.75	59.21	88.12	**54.56**	**36.54**	**23.66**	59.23	46.07	21.30	47.91	13.69	**65.80**	**79.78**	52.52
1-8	**Baseline**	55.23	98.76	**63.17**	63.14	58.60	87.80	54.09	35.82	18.94	62.32	**48.17**	33.26	44.97	8.46	64.79	82.16	**59.32**
	LSST [22]	35.45	93.53	2.97	39.60	51.74	64.31	31.32	28.72	6.25	38.01	19.35	1.51	32.42	20.83	47.26	60.00	29.46
	ClassHyPer [15]	43.78	98.42	55.84	46.07	35.74	87.45	46.80	12.55	11.91	57.52	45.92	2.45	37.60	11.14	29.81	73.03	48.26
	ST++ [21]	54.62	98.73	59.73	63.91	63.93	87.67	**57.04**	31.77	**19.56**	**62.37**	47.57	15.77	**58.30**	10.81	58.17	81.58	57.05
	DAST(Ours)	**58.09**	**98.82**	61.91	**67.53**	**67.81**	**90.09**	54.19	**42.34**	12.02	61.40	45.98	**34.02**	50.08	**36.17**	**68.11**	**83.54**	55.42
1-4	**Baseline**	59.56	98.86	64.84	67.89	**70.68**	88.76	58.23	39.27	28.60	65.17	**50.38**	23.89	**54.47**	40.54	63.91	81.82	55.62
	LSST [22]	36.19	94.86	6.09	43.45	43.79	74.837	29.84	31.20	0.07	43.00	18.89	0.00	30.32	31.19	46.24	56.90	28.41
	ClassHyPer [15]	43.46	98.31	52.14	49.09	30.19	88.06	48.59	21.74	19.68	54.95	41.25	1.67	35.00	12.28	35.31	57.72	49.39
	ST++ [21]	57.74	98.74	61.56	66.01	69.60	89.71	55.11	40.81	25.84	62.21	43.95	13.73	50.37	**57.33**	52.12	80.73	55.99
	DAST(Ours)	**62.05**	**98.94**	**68.60**	**68.37**	70.44	**90.33**	**62.54**	**48.34**	**33.24**	**65.30**	47.55	**24.25**	52.49	51.76	**68.26**	**83.51**	**58.93**

achievement is observed in the Soccer Ball Field category, where our approach surpasses ST++ [21] by an impressive 26.02%.

Moving to the 1/8 labeled data scenario, our approach again shines, yielding a 2.86% improvement in mean IoU compared to the baseline. Notably, all other advanced methods lag behind the baseline in this setting. It is important to highlight that the supervised baseline's performance in the Round category is significantly deficient, standing at just 8.46%. This is primarily due to the random selection of labeled data, which includes rare Round category samples. Nevertheless, our approach achieves a remarkable 36.17%, outperforming the baseline by a significant margin of 27.71%. This result confirms our approach's efficiency and shows that it can perform well in categories where sample data is scarce.

In the 1/4 labeled data scenario, our approach again shines with a 2.49% improvement in mean IoU than the baseline. In contrast, other advanced methods continue to underperform relative to the baseline. It is worth mentioning that the results of LSST [22] in our experiments differ from their reported values, primarily due to differences in data preparation. LSST [22] resized images for training and testing while we employed cropping. Consequently, the difficulty level of our task was higher, and our mIoU in the 1/4 labeled data scenario (62.05%) aligns with their reported performance in 300 labeled images, representing almost 1/4 of the labeled data. In contrast, their method achieved an mIoU of only 36.19% with cropped data, a substantial 25.86% lower than our result. These outcomes affirm the robust learning capabilities of our method, which effectively leverages labeled and unlabeled data, exhibiting resilience even in the presence of a large volume of unlabeled data.

Our method displays its most significant advantages in scenarios with limited labeled data. The most substantial improvement, a 3.19% increase in mIoU, occurs in the 1/32 labeled data scenario. This underscores our method's effectiveness in enhancing segmentation model performance, even when labeled data is scarce. Moreover, our results outperform previous methods, affirming our method's ability to harness unlabeled images effectively to enhance segmentation performance, especially when the amount of labeled data is small.

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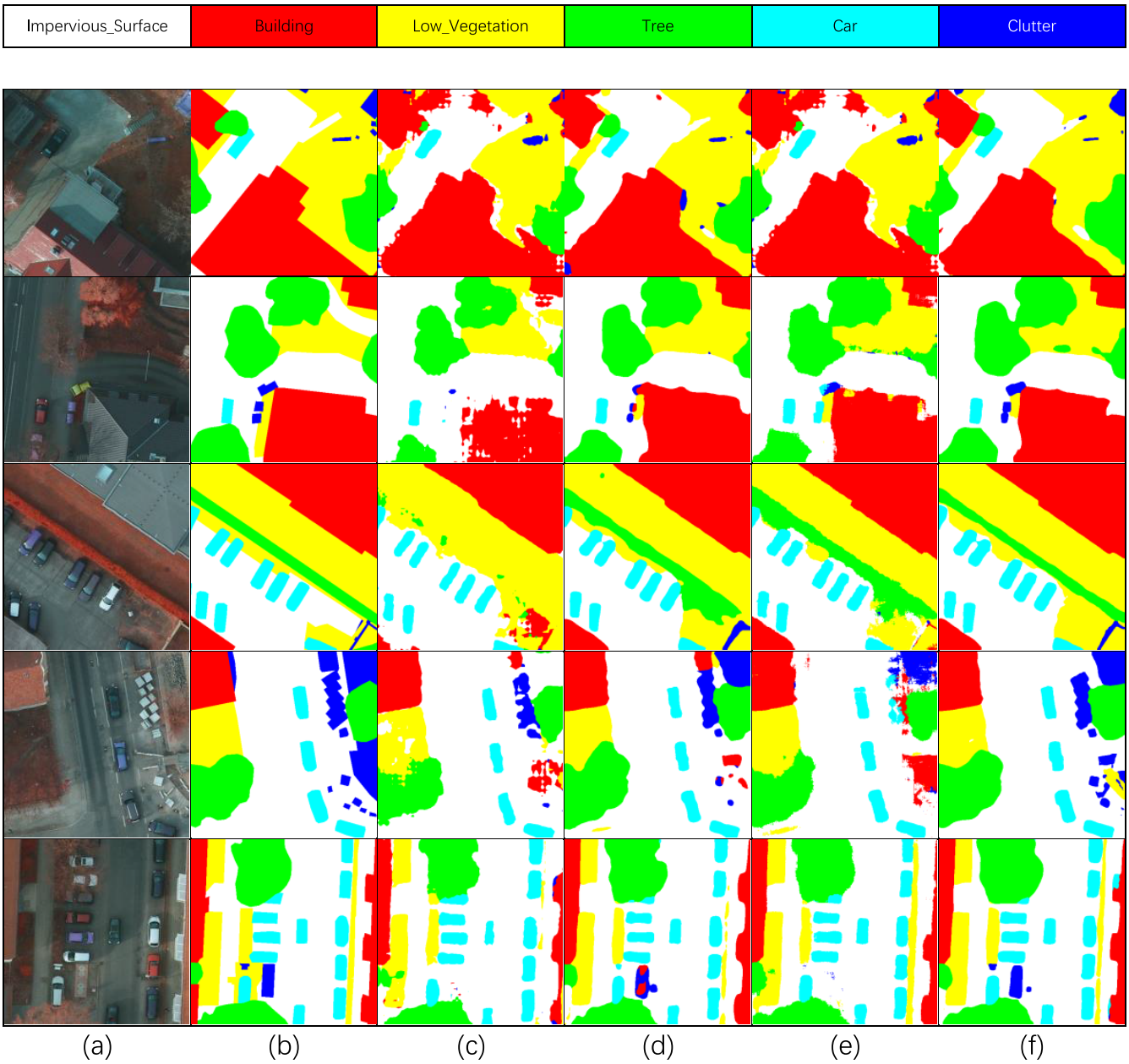


Fig. 5. Visual comparison of semantic segmentation results using various semi-supervised methods on the Potsdam test dataset with a 1/32 labeled data scenario. (a) Input image. (b) Ground truth. (c) LSST. (d) ST++. (e) ClassHyper. (f) DAST(Ours).

F. Visual Comparison

1) Visual Comparison on POTSDAM: Fig. 5 presents a visual comparison of the semantic segmentation results, providing clear insights into the strengths of our methodology. Our approach demonstrates a significant advantage in accurately identifying and segmenting the clutter class (depicted in blue). In the bottom row of images, it is evident that the predictions produced by LSST [22] and ClassHyper [15] fail to detect the clutter class entirely. In contrast, ST++ [21] manages to identify a portion of the clutter but erroneously predicts some clutter areas as buildings. Conversely, our method's predictions detect nearly all parts of the clutter class with minimal redundancy. It is worth highlighting that our method yields predictions without scattered points, a distinct characteristic that sets it apart from other techniques. This differentiation is attributed to our unique approach during pseudo-label sampling, where we group pixels into region-based selections. Leveraging correlations among neighboring pixels has significantly enhanced our model's performance.

2) Visual Comparison on iSAID: Fig. 6 provides a visual comparison of the semantic segmentation results, offering clear insights into the strengths of our methodology. In the first three rows, we can observe that ClassHyper [15] achieves similar performance to our method. However, in the fourth row, the performance of our method in the roundabout class is notably better than that of ClassHyper [15]. ClassHyper [15] cannot detect the entire plane in the fifth row. Compared to ST++, we can see from the third row that our method's prediction performance for the harbor class is notably superior

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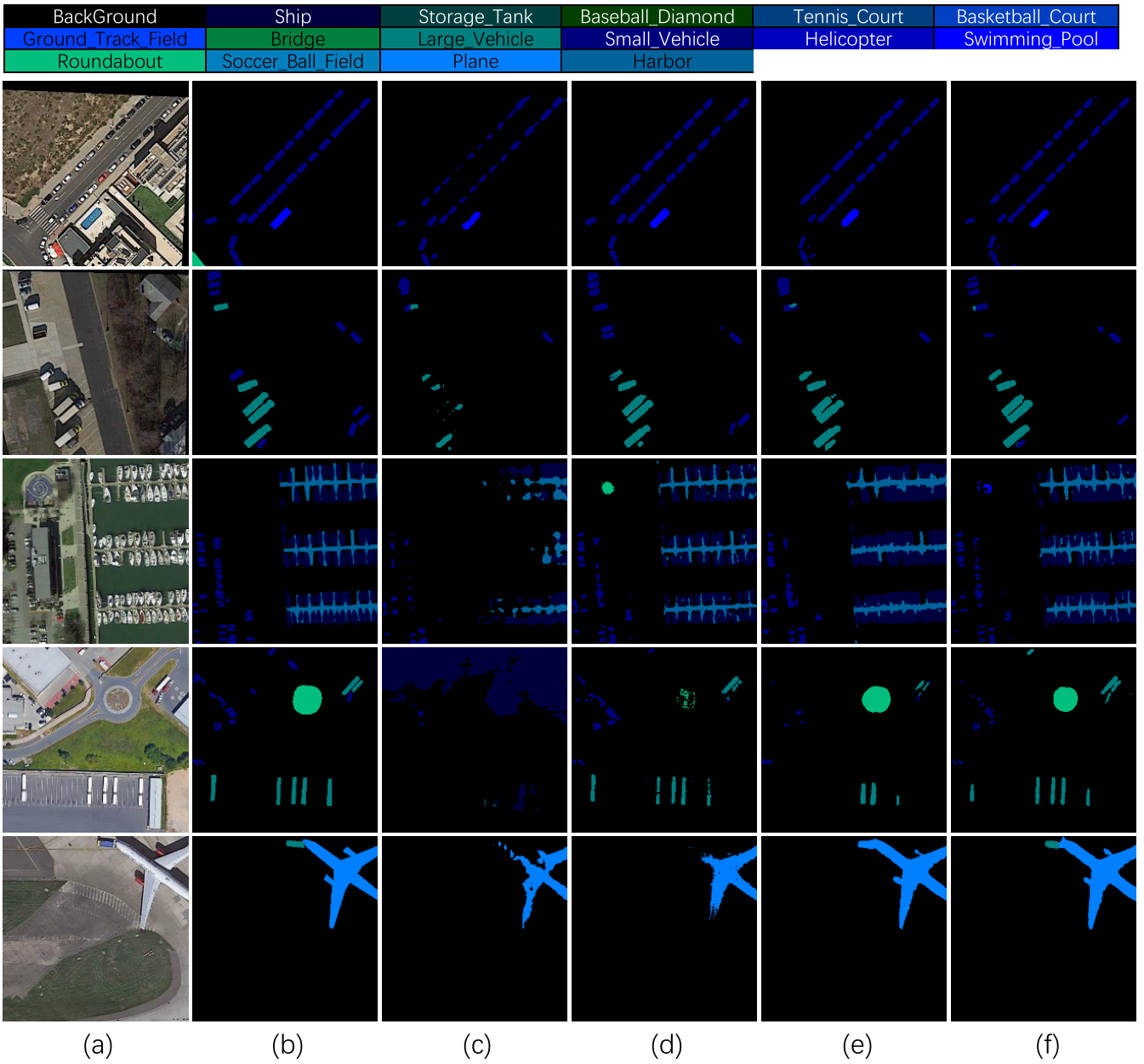


Fig. 6. Visual comparison of semantic segmentation results using various semi-supervised methods on the iSAID test dataset with a 1/32 labeled data scenario. (a) Input image. (b) Ground truth. (c) LSST. (d) ClassHyper. (e)ST++. (f) DAST(Ours).

to ST++ [21]. In the first and second rows, ST++ [21] misses some small vehicles, and in the fourth row, it misses some large vehicles. In the fifth row, ST++ [21] mistakenly regards a large vehicle as a part of a plane. Collectively, these visual comparisons illustrate the superior accuracy of our approach compared to other methods, providing an intuitive depiction of its performance.

G. Ablation Study

Our method is composed of DPS based on class distribution, DM, and ATU. In this section, we thoroughly assess the exact benefits of the three modules. The ablation study is conducted in the same configuration as the above experiment on the Potsdam dataset under 1/32 labeled data.

Table V shows that compared to baseline, the DPS module improves the mean IoU by 2.57%, the DPS module combined with the DM module improves the mean IoU by 3.83%, the DPS module combined with the ATU module improves the mean IoU by 3.79%, and three modules improves the mean IoU by 5.46%. Each module improves the mIoU by 2.57%,1.26%, and 1.63% separately. It is worth noticing that the DPS module shows the most significant improvement as a single module, mainly because it can update the pseudo-labels with training and decrease the error rate of pseudo-labels, preventing the model from learning the error information. The DM module achieves mIoU improvement of 1.34%, which indicates that it is significant to match the distribution of unlabeled data and labeled data to push the model to learn accurate data distribution. The ATU module achieves mIoU

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TABLE V
ABLATION STUDY ON 1–32 POTSDAM

DPS	DM	ATU	mIoU
			66.37
✓			68.94
✓		✓	70.16
✓	✓		70.20
✓	✓	✓	71.83

improvement of 1.63%, which shows that adaptive threshold with training can enhance the quality of pseudo-labels, reducing false positives and increasing accuracy.

V. CONCLUSION

This article introduces a novel semi-supervised semantic segmentation approach designed specifically for RS images. Our method, rooted in self-training and CR, unfolds in a two-stage process. Initially, we employ supervised training using labeled data, subsequently utilizing the trained model to predict pseudo-labels for unlabeled data. This step effectively addresses the class imbalance challenge through a careful sampling process. We introduce a distribution adjustment strategy to compute the CR loss in the retraining phase. The core objective of this strategy is to bring the model's distribution closer to the actual distribution of the data. Concurrently, it adapts the threshold for different classes, enhancing the quality of labels and the overall utility of the data. By amalgamating self-training, CR, and specialized strategies tailored for RS imagery, our method achieves state-of-the-art performance on both the Potsdam and iSAID datasets. This accomplishment demonstrates that our approach harnesses the advantages of SSL, ultimately amplifying the efficacy of semantic segmentation in RS applications.

REFERENCES

[1] C. Persello et al., "Deep learning and Earth observation to support the sustainable development goals: Current approaches, open challenges, and future opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 172–200, Jun. 2022.

[2] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[3] J. Li, S. Zi, R. Song, Y. Li, Y. Hu, and Q. Du, "A stepwise domain adaptive segmentation network with covariate shift alleviation for remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618515.

[4] N. He, L. Fang, and A. Plaza, "Hybrid first and second order attention UNet for building segmentation in remote sensing images," *Sci. China Inf. Sci.*, vol. 63, no. 4, pp. 1–12, Apr. 2020.

[5] Y. Li, T. Shi, Y. Zhang, W. Chen, Z. Wang, and H. Li, "Learning deep semantic segmentation network under multiple weakly-supervised constraints for cross-domain remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 20–33, May 2021.

[6] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[7] C. Peng, K. Zhang, Y. Ma, and J. Ma, "Cross fusion net: A fast semantic segmentation network for small-scale semantic information capturing in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601313.

[8] X. Pan, L. Gao, A. Marinoni, B. Zhang, F. Yang, and P. Gamba, "Semantic labeling of high resolution aerial imagery and LiDAR data with fine segmentation network," *Remote Sens.*, vol. 10, no. 5, p. 743, May 2018.

[9] X. Pan, L. Gao, B. Zhang, F. Yang, and W. Liao, "High-resolution aerial imagery semantic labeling with dense pyramid network," *Sensors*, vol. 18, no. 11, p. 3774, Nov. 2018.

[10] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, 2022.

[11] L. Sun, S. Cheng, Y. Zheng, Z. Wu, and J. Zhang, "SPANet: Successive pooling attention network for semantic segmentation of remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4045–4057, 2022.

[12] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[13] Y. Ouali, C. Hudelot, and M. Tami, "Semi-supervised semantic segmentation with cross-consistency training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12674–12684.

[14] M. Cordts et al., "The cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3213–3223.

[15] Y. He, J. Wang, C. Liao, B. Shan, and X. Zhou, "ClassHyPer: ClassMix-based hybrid perturbations for deep semi-supervised semantic segmentation of remote sensing imagery," *Remote Sens.*, vol. 14, no. 4, p. 879, Feb. 2022.

[16] J. Li, B. Sun, S. Li, and X. Kang, "Semisupervised semantic segmentation of remote sensing images with consistency self-training," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5615811.

[17] T. Miyato, S.-I. Maeda, M. Koyama, and S. Ishii, "Virtual adversarial training: A regularization method for supervised and semi-supervised learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 8, pp. 1979–1993, Aug. 2018.

[18] J. Kang, Z. Wang, R. Zhu, X. Sun, R. Fernandez-Beltran, and A. Plaza, "PiCoCo: Pixelwise contrast and consistency learning for semisupervised building footprint segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10548–10559, 2021.

[19] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.

[20] D.-H. Lee et al., "Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks," in *Proc. Workshop Challenges Represent. Learn.*, Atlanta, GA, USA, 2013, vol. 3, no. 2, p. 896.

[21] L. Yang, W. Zhuo, L. Qi, Y. Shi, and Y. Gao, "ST++: Make self-training work better for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 4268–4277.

[22] X. Lu et al., "Simple and efficient: A semisupervised learning framework for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5543516.

[23] Y. Wu, G. Mu, C. Qin, Q. Miao, W. Ma, and X. Zhang, "Semi-supervised hyperspectral image classification via spatial-regulated self-training," *Remote Sens.*, vol. 12, no. 1, p. 159, Jan. 2020.

[24] J. Wang, C. H. Q. Ding, S. Chen, C. He, and B. Luo, "Semi-supervised remote sensing image semantic segmentation via consistency regularization and average update of pseudo-label," *Remote Sens.*, vol. 12, no. 21, p. 3603, Nov. 2020.

[25] J.-X. Wang, S.-B. Chen, C. H. Q. Ding, J. Tang, and B. Luo, "RanPaste: Paste consistency and pseudo label for semisupervised remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 2002916.

[26] R. Dong, X. Pan, and F. Li, "DenseU-Net-based semantic segmentation of small objects in urban remote sensing images," *IEEE Access*, vol. 7, pp. 65347–65356, 2019.

[27] D. Berthelot et al., "ReMixMatch: Semi-supervised learning with distribution alignment and augmentation anchoring," 2019, *arXiv:1911.09785*.

[28] L. Zhang, L. Yang, T. Ma, F. Shen, Y. Cai, and C. Zhou, "A self-training semi-supervised machine learning method for predictive mapping of soil classes with limited sample data," *Geoderma*, vol. 384, Feb. 2021, Art. no. 114809.

[29] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2022.

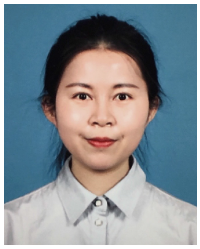
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[30] N. Parmar et al., "Image transformer," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 4055–4064.
[31] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[32] H. Wang, C. Tao, J. Qi, R. Xiao, and H. Li, "Avoiding negative transfer for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4413215.
[33] Z. Xu, W. Zhang, T. Zhang, Z. Yang, and J. Li, "Efficient transformer for remote sensing image segmentation," *Remote Sens.*, vol. 13, no. 18, p. 3585, Sep. 2021.
[34] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.
[35] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.
[36] S. Mittal, M. Tatarchenko, and T. Brox, "Semi-supervised semantic segmentation with high-and low-level consistency," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1369–1379, Apr. 2019.
[37] Z. Feng et al., "DMT: Dynamic mutual training for semi-supervised learning," *Pattern Recognit.*, vol. 130, Oct. 2022, Art. no. 108777.
[38] R. Ke, A. I. Aviles-Rivero, S. Pandey, S. Reddy, and C.-B. Schönlieb, "A three-stage self-training framework for semi-supervised semantic segmentation," *IEEE Trans. Image Process.*, vol. 31, pp. 1805–1815, 2022.
[39] Y. Liu, Y. Tian, Y. Chen, F. Liu, V. Belagiannis, and G. Carneiro, "Perturbed and strict mean teachers for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4258–4267.
[40] Y. Wang et al., "Semi-supervised semantic segmentation using unreliable pseudo-labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 4248–4257.
[41] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2020, pp. 596–608.
[42] Y. Wang et al., "FreeMatch: Self-adaptive thresholding for semi-supervised learning," 2022, *arXiv:2205.07246*.
[43] H. Chen et al., "SoftMatch: Addressing the quantity-quality trade-off in semi-supervised learning," 2023, *arXiv:2301.10921*.
[44] B. Kim, J. Choo, Y.-D. Kwon, S. Joe, S. Min, and Y. Gwon, "SelfMatch: Combining contrastive self-supervision and consistency for semi-supervised learning," 2021, *arXiv:2101.06480*.
[45] I. J. Goodfellow et al., "Generative adversarial networks," Tech. Rep., 2014.
[46] F. Rottensteiner et al., "THE ISPRS benchmark on urban object classification and 3D building reconstruction," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 3, pp. 293–298, Jul. 2012.
[47] S. W. Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2019, pp. 28–37.
[48] G. Csurka, D. Larlus, and F. Perronnin, "What is a good evaluation measure for semantic segmentation?" in *Proc. BMVC*, 2013.
[49] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 418–434.
[50] H. Zhang et al., "Context encoding for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Mar. 2018, pp. 7151–7160.



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